

ESG360

Smart BRSR Reporting & ESG Compliance Portal

Project Type	Web-Based ESG and BRSR Reporting Software
Problem Statement	Development of an online portal for BRSR reporting of Corporate ESG Reports.
Frontend	React.js
Backend	Node.js + Express.js
Database	MongoDB
Storage	Cloudinary
Auth	JWT
API	REST API

1. PROJECT OBJECTIVE

The objective of ESG360 is to develop a centralized web-based platform that helps organizations collect, validate, manage, consolidate, analyze and report ESG (Environmental, Social and Governance) information.

The system will support reporting at multiple levels:

Group → Subsidiary → Business Unit → Project

2. PROJECT USERS

1. Super Admin
2. Group ESG Admin
3. Subsidiary Admin
4. Business Unit Manager
5. Project/Department User
6. ESG Manager
7. Compliance Officer
8. Management
9. Auditor/Reviewer

3. MAIN FUNCTIONAL MODULES

3.1 Authentication

- Register
- Login
- Logout
- JWT authentication
- Role-based access

3.2 Organization Management

- Group management
- Subsidiary management
- Business Unit management

- Project management

3.3 ESG Data Collection

- Environmental data
- Social data
- Governance data
- Reporting period
- Units and values
- Data source

3.4 Environmental Module

- Energy consumption
- Electricity consumption
- Fuel consumption
- Water consumption
- Waste
- Emissions
- Renewable energy
- Environmental incidents

3.5 Social Module

- Employee information
- Employee diversity
- Training
- Health and safety
- Workplace accidents
- Employee welfare
- Community development
- Grievances

3.6 Governance Module

- Board information
- Policies
- Ethics and compliance
- Anti-corruption
- Whistleblower complaints
- Risk management
- Regulatory compliance
- Data privacy

3.7 BRSR Module

- BRSR questions/indicators
- Data mapping
- Completion tracking

- BRSR-oriented report preparation

3.8 Data Validation

- Required field validation
- Data type validation
- Unit validation
- Duplicate checking
- Missing data detection
- Correction workflow

3.9 Document Management

- Upload supporting documents
- Store document information
- Link documents with ESG data
- Delete documents
- Cloudinary integration

3.10 Approval Workflow

- Draft → Submitted → Under Review → Validated → Approved
- If incorrect: Under Review → Correction Required → Resubmission

3.11 Dashboard

- ESG KPIs
- Reporting completion
- Validation status
- Pending submissions
- Environmental metrics
- Social metrics
- Governance metrics

3.12 Analytics

- Year-wise trends
- Entity-wise analysis
- Project-wise analysis
- ESG performance analysis

3.13 Reports

- ESG reports
- BRSR-oriented reports
- Entity reports
- Consolidated group reports

3.14 Notifications

- Pending submission
- Correction required
- Approval notification

- Reporting deadline

3.15 Audit Logs

- User activity
- Data changes
- Approval history
- Timestamp
- Previous and updated values

4. FRONTEND DEVELOPMENT PLAN

Phase 1: React Project Setup

- Create React application
- Install required packages
- Configure routing
- Create global styling

Phase 2: Common Components

- Navbar
- Sidebar
- Footer
- StatCard
- ChartCard
- DataTable

Phase 3: Authentication Pages

- Landing page
- Login page
- Register page

Phase 4: Dashboard

- KPI cards
- ESG overview
- Charts
- Reporting status
- Recent activities

Phase 5: Organization

- Group
- Subsidiary
- Business Unit
- Project

Phase 6: ESG Modules

- Environmental
- Social

- Governance

Phase 7: BRSR

- BRSR indicators
- Data collection
- Completion status

Phase 8: Data Management

- Data Collection
- Validation
- Documents
- Approvals

Phase 9: Reporting

- Reports
- Analytics

Phase 10: Administration

- Notifications
- Audit Logs

5. BACKEND DEVELOPMENT PLAN

Phase 1: Server Setup

- Initialize Node.js project
- Install Express
- Configure server
- Configure environment variables

Phase 2: Database

- Connect MongoDB
- Create database models
- Configure database connection

Phase 3: Authentication

- User registration
- Login
- JWT token
- Password security
- Role-based authorization

Phase 4: Organization APIs

- Organization API
- Subsidiary API
- Business Unit API
- Project API

Phase 5: ESG APIs

- Create ESG data
- Read ESG data
- Update ESG data
- Delete ESG data
- Filter ESG data

Phase 6: BRSR APIs

- BRSR indicators
- BRSR data
- Reporting status
- Report preparation

Phase 7: Validation

- Validate ESG values
- Check missing fields
- Check incorrect values
- Return validation messages

Phase 8: Document Management

- File upload
- Cloudinary storage
- Document metadata
- Document deletion

Phase 9: Approval Workflow

- Submit
- Review
- Approve
- Reject
- Correction required

Phase 10: Reporting

- ESG report
- BRSR report
- Consolidated report
- Report export

Phase 11: Audit and Notification

- Audit logs
- Notifications
- Activity tracking

6. DATABASE MODELS

User

- name

- email
- password
- role
- organization
- status

Organization

- name
- type
- parentOrganization
- location
- status

ESGData

- category
- indicator
- value
- unit
- reportingYear
- organization
- project
- status
- submittedBy

BRSRReport

- organization
- reportingYear
- indicators
- status
- generatedBy

Document

- fileName
- fileUrl
- documentType
- uploadedBy
- relatedESGData

Notification

- user
- title
- message
- status

AuditLog

- user
- action
- module
- oldValue
- newValue
- timestamp

7. API STRUCTURE

Authentication

Method	Endpoint
POST	/api/auth/register
POST	/api/auth/login
GET	/api/auth/profile

ESG

Method	Endpoint
GET	/api/esg
POST	/api/esg
GET	/api/esg/:id
PUT	/api/esg/:id
DELETE	/api/esg/:id

BRSR

Method	Endpoint
GET	/api/brsr
POST	/api/brsr
GET	/api/brsr/:id
PUT	/api/brsr/:id

Reports

Method	Endpoint
GET	/api/reports
POST	/api/reports/generate

Documents

Method	Endpoint
POST	/api/documents/upload
GET	/api/documents
DELETE	/api/documents/:id

8. DATA FLOW

Project / Department User → ESG Data Entry → Validation → Submit → Business Unit Review →
Subsidiary Validation → Group-Level Consolidation → ESG Dashboard → BRSR-Oriented Reporting

9. SECURITY PLAN

- JWT authentication
- Password hashing
- Role-based access control
- Protected API routes
- Input validation
- Secure file upload
- Environment variables
- Audit logging
- Access restrictions by organization

10. TESTING PLAN

Frontend

- Component testing
- Page testing
- Form validation
- Responsive testing

Backend

- API testing
- Authentication testing
- Database testing
- Validation testing
- File upload testing

Integration

- Frontend ↔ Backend
- Backend ↔ MongoDB
- Backend ↔ Cloudinary

11. DEPLOYMENT PLAN

Frontend → React Deployment

Backend → Node.js / Express Deployment

Database → MongoDB

File Storage → Cloudinary

12. DEVELOPMENT ORDER

Step 1 → Project setup

Step 2 → Frontend structure

Step 3 → Backend structure

Step 4 → MongoDB connection

Step 5 → Authentication

Step 6 → Organization management

Step 7 → ESG data collection

Step 8 → Environmental module

Step 9 → Social module

Step 10 → Governance module

Step 11 → BRSR module

Step 12 → Validation

Step 13 → Document management

Step 14 → Approval workflow

Step 15 → Dashboard

Step 16 → Analytics

Step 17 → Reports

Step 18 → Notifications

Step 19 → Audit logs

Step 20 → Testing

Step 21 → Deployment

13. FINAL PROJECT FLOW

ESG360



■■■■ FRONTEND – React.js



■■■■ User Interface



■■■■ BACKEND – Node + Express



■■■■ REST APIs



MongoDB



ESG DATA



VALIDATION



APPROVAL



CONSOLIDATION



ESG ANALYTICS



BRSR-ORIENTED REPORT

ESG360 — Smart BRSR Reporting & ESG Compliance Portal